

SKILL WEEK – 1

Session – 1

TASK 1: To Install Linux VM, Configure GCC, Setup Git Repository, Create Project Structure, Understand Shell Architecture, Build Initial Makefile

```
keerthi_priya@LAPTOP-OSGVCE7A:~$ gcc --version
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0
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This is free software; see the source for copying conditions. There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

keerthi_priya@LAPTOP-OSGVCE7A:~$ git --version
git version 2.53.0
keerthi_priya@LAPTOP-OSGVCE7A:~$ cd
keerthi_priya@LAPTOP-OSGVCE7A:~$ touch file.txt
```

TASK 2: To Analyze process abstraction, execute fork(), understand exec() family, analyze parent-child relationships, inspect process tree, practice system call tracing

```
GNU nano 8.7.1
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t pid;

    printf("Parent Process Started\n");
    printf("Parent PID: %d\n", getpid());

    pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {
        printf("\nChild Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());
        printf("Executing ls using exec()\n");

        execlp("ls", "ls", "-l", NULL);

        perror("exec failed");
        exit(1);
    } else {
        printf("\nParent Process\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);
        printf("Parent waiting for child...\n");

        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}
```

```

keerthi_priya@LAPTOP-OSGV x + v
keerthi_priya@LAPTOP-OSGVCE7A:~$ gcc skill_week1_task2.c -o skill_week1_task2
keerthi_priya@LAPTOP-OSGVCE7A:~$ ./skill_week1_task2
Parent Process Started
Parent PID: 553

Parent Process
Parent PID: 553
Child PID: 554
Parent waiting for child...

Child Process
Child PID: 554
Parent PID: 553
Executing ls using exec()
total 644
prw-r--r-- 1 keerthi_priya keerthi_priya 0 Aug 29 08:23 ' info'
drwxr-xr-x 11 keerthi_priya keerthi_priya 4096 Aug 1 08:15 OSSP
drwxr-xr-x 2 keerthi_priya keerthi_priya 4096 Aug 3 16:41 OS_LAB
drwxr-xr-x 2 keerthi_priya keerthi_priya 4096 Aug 3 04:46 ProcessTask
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16344 Sep 17 08:32 a.out
prw-r--r-- 1 keerthi_priya keerthi_priya 0 Sep 17 08:05 client_to_server
-rw-r--r-- 1 keerthi_priya keerthi_priya 70 Sep 18 04:10 error.txt
-rw-r--r-- 1 keerthi_priya keerthi_priya 0 Aug 3 15:09 file.txt
-rw-r--r-- 1 keerthi_priya keerthi_priya 0 Aug 29 08:34 info.txt
-rwxr-xr-x 1 keerthi_priya keerthi_priya 34 Sep 18 03:41 input.txt
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16288 Sep 17 12:50 memory_analysis
-rw-r--r-- 1 keerthi_priya keerthi_priya 853 Sep 17 12:47 memory_analysis..save
-rw-r--r-- 1 keerthi_priya keerthi_priya 853 Sep 17 12:50 memory_analysis.c
-rw-r--r-- 1 keerthi_priya keerthi_priya 51 Aug 3 15:21 numbers.txt
-rw-r--r-- 1 keerthi_priya keerthi_priya 87 Sep 18 05:13 output.txt
-rw-r--r-- 1 keerthi_priya keerthi_priya 657 Sep 17 08:12 p-6.c
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16344 Sep 17 08:33 p62
drwxr-xr-x 2 keerthi_priya keerthi_priya 4096 Aug 22 06:00 practical3
drwxr-xr-x 2 keerthi_priya keerthi_priya 4096 Aug 17 05:34 practical4
drwxr-xr-x 2 keerthi_priya keerthi_priya 4096 Aug 26 10:08 practical5
drwxr-xr-x 1 keerthi_priya keerthi_priya 4096 Sep 17 13:11 practical_memory
prw-r--r-- 1 keerthi_priya keerthi_priya 0 Sep 17 08:05 server_to_client
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16312 Sep 17 17:36 session1_task1
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16312 Sep 17 17:37 session10_task1
-rw-r--r-- 1 keerthi_priya keerthi_priya 1332 Sep 17 17:37 session10_task1.c
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16568 Sep 17 17:44 session10_task2
-rw-r--r-- 1 keerthi_priya keerthi_priya 2273 Sep 17 17:44 session10_task2.c
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16352 Sep 18 03:19 session11_task1
-rw-r--r-- 1 keerthi_priya keerthi_priya 1747 Sep 18 02:58 session11_task1.c
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16648 Sep 18 03:11 session11_task2
-rw-r--r-- 1 keerthi_priya keerthi_priya 3559 Sep 18 03:08 session11_task2.c
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16352 Sep 18 03:20 session12_task1
-rw-r--r-- 1 keerthi_priya keerthi_priya 1587 Sep 18 03:20 session12_task1.c
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16656 Sep 18 04:08 session12_task2
-rw-r--r-- 1 keerthi_priya keerthi_priya 3580 Sep 18 03:26 session12_task2.c

```

```

keerthi_priya@LAPTOP-OSGV x + v
-rw-r--r-- 1 keerthi_priya keerthi_priya 1332 Sep 17 17:37 session10_task1.c
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16568 Sep 17 17:44 session10_task2
-rw-r--r-- 1 keerthi_priya keerthi_priya 2273 Sep 17 17:44 session10_task2.c
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16352 Sep 18 03:19 session11_task1
-rw-r--r-- 1 keerthi_priya keerthi_priya 1747 Sep 18 02:58 session11_task1.c
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16648 Sep 18 03:11 session11_task2
-rw-r--r-- 1 keerthi_priya keerthi_priya 3559 Sep 18 03:08 session11_task2.c
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16352 Sep 18 03:20 session12_task1
-rw-r--r-- 1 keerthi_priya keerthi_priya 1587 Sep 18 03:20 session12_task1.c
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16656 Sep 18 04:08 session12_task2
-rw-r--r-- 1 keerthi_priya keerthi_priya 3580 Sep 18 03:26 session12_task2.c
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16608 Sep 18 03:41 session13_task1
-rw-r--r-- 1 keerthi_priya keerthi_priya 2144 Sep 18 03:40 session13_task1.c
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16656 Sep 18 03:54 session13_task2
-rw-r--r-- 1 keerthi_priya keerthi_priya 2653 Sep 18 03:54 session13_task2.c
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16656 Sep 18 04:02 session14_task1
-rw-r--r-- 1 keerthi_priya keerthi_priya 2607 Sep 18 04:02 session14_task1.c
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16656 Sep 18 04:09 session14_task2
-rw-r--r-- 1 keerthi_priya keerthi_priya 2714 Sep 18 04:09 session14_task2.c
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16744 Sep 18 05:13 session15_task1
-rw-r--r-- 1 keerthi_priya keerthi_priya 3510 Sep 18 05:12 session15_task1.c
-rw-r--r-- 1 keerthi_priya keerthi_priya 3919 Sep 18 05:06 session15_task1.c.save
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16808 Sep 18 07:43 session15_task2
-rw-r--r-- 1 keerthi_priya keerthi_priya 5807 Sep 18 07:43 session15_task2.c
-rw-r--r-- 1 keerthi_priya keerthi_priya 5805 Sep 18 05:19 session15_task2.c.save
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16768 Sep 18 08:01 session16_task1
-rw-r--r-- 1 keerthi_priya keerthi_priya 3987 Sep 18 08:01 session16_task1.c
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16992 Sep 18 08:05 session16_task2
-rw-r--r-- 1 keerthi_priya keerthi_priya 5832 Sep 18 08:04 session16_task2.c
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16664 Sep 17 14:17 session3_task1
-rw-r--r-- 1 keerthi_priya keerthi_priya 4638 Sep 17 14:17 session3_task1.c
-rw-r--r-- 1 keerthi_priya keerthi_priya 2957 Sep 17 15:18 session3_task1.c.save
drwxr-xr-x 2 keerthi_priya keerthi_priya 4096 Sep 17 14:33 session3_task2.c
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16312 Sep 17 17:01 session8_task1
-rw-r--r-- 1 keerthi_priya keerthi_priya 2875 Sep 17 17:00 session8_task1.c
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16736 Sep 17 17:09 session8_task2
-rw-r--r-- 1 keerthi_priya keerthi_priya 2005 Sep 17 17:09 session8_task2.c
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16400 Sep 17 17:24 session9_task1
-rw-r--r-- 1 keerthi_priya keerthi_priya 1619 Sep 17 17:24 session9_task1.c
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16656 Sep 17 17:30 session9_task2
-rw-r--r-- 1 keerthi_priya keerthi_priya 1952 Sep 17 17:30 session9_task2.c
drwxr-xr-x 2 keerthi_priya keerthi_priya 4096 Aug 20 06:35 skill1
drwxr-xr-x 3 keerthi_priya keerthi_priya 4096 Aug 22 06:46 skill4
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16304 Sep 19 04:44 skill_week1_task2
-rw-r--r-- 1 keerthi_priya keerthi_priya 874 Sep 19 04:44 skill_week1_task2.c
-rwxr-xr-x 1 keerthi_priya keerthi_priya 16296 Aug 26 10:10 task2
-rw-r--r-- 1 keerthi_priya keerthi_priya 1015 Aug 26 10:09 task2.c
prw-r--r-- 1 keerthi_priya keerthi_priya 0 Aug 29 08:23 tmp
drwxr-xr-x 2 keerthi_priya keerthi_priya 4096 Aug 12 08:30 week3
Child process completed.

```

Session – 2

TASK 1: To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

```
keerthi_priya@LAPTOP-OSGV  ×  +  v
GNU nano 8.7.1
#include <stdio.h>
#include <string.h>

int main() {
    char command[100];

    printf("OSSP Interactive Shell\n");

    while (1) {
        printf("myshell> ");

        if (fgets(command, sizeof(command), stdin) == NULL) {
            break;
        }

        command[strcspn(command, "\n")] = '\0';

        if (strcmp(command, "exit") == 0) {
            printf("Exiting shell...\n");
            break;
        }

        if (strlen(command) == 0) {
            continue;
        }

        printf("You entered: %s\n", command);
    }

    return 0;
}
```

```
keerthi_priya@LAPTOP-OSGVCE7A:~$ nano session2_task1.c
keerthi_priya@LAPTOP-OSGVCE7A:~$ gcc session2_task1.c -o session2_task1
keerthi_priya@LAPTOP-OSGVCE7A:~$ ./session2_task1
OSSP Interactive Shell
myshell> hello
You entered: hello
myshell> test
You entered: test
myshell> exit
Exiting shell...
keerthi_priya@LAPTOP-OSGVCE7A:~$ |
```

TASK 2: To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction

```
keerthi_priya@LAPTOP-OSGV  ×  +  v
GNU nano 8.7.1
#include <stdio.h>

int main() {
    char buffer[100];
    int ch;
    int index = 0;

    printf("Type a command and press Enter:\n");
    printf("input> ");

    while ((ch = getchar()) != '\n' && ch != EOF) {

        // Handle Backspace
        if (ch == 127 || ch == 8) {
            if (index > 0) {
                index--;
            }
        }
        // Store characters in buffer
        else if (index < 99) {
            buffer[index++] = ch;
        }

        // End the string
        buffer[index] = '\0';

        printf("\nCommand entered: %s\n", buffer);
        printf("Characters stored: %d\n", index);

        return 0;
    }
}
```

```
keerthi_priya@LAPTOP-OSGVCE7A:~$ nano session2_task2.c
keerthi_priya@LAPTOP-OSGVCE7A:~$ gcc session2_task2.c -o session2_task2
keerthi_priya@LAPTOP-OSGVCE7A:~$ ./session2_task2
Type a command and press Enter:
input> hello world

Command entered: hello world
Characters stored: 11
keerthi_priya@LAPTOP-OSGVCE7A:~$ |
```